

15A	CSE421: Computer Networks Quiz #1 Time: 15 Minutes	Marks:	/15
Serial:	Name:	ID:	

Figures in the right margin indicate marks.

- Q1** Alice uploads a photo to a social media website. The upload is successful. Below is a list of steps that Alice's PC took to send the upload request to the server. Mention the OSI layers associated with each of these tasks and re-arrange them in the correct order of the OSI model. 7
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- A. The destination physical address of the next-hop device is determined.
 - B. Alice selects the photo and clicks "Upload."
 - C. The source and destination address of the datagram is determined.
 - D. The frames are converted to bits and sent through the network interface card.
 - E. A session is established between Alice's PC and the web server.
 - F. The image file is compressed and encrypted before transmission.
 - G. The browser application identifies the upload process.
- Q2** Which TCP/IP layer is concerned with the **Port address**, and what is the purpose of using this address 3
- Q3** Identify the purpose of **bit synchronization** in the **physical layer**. 3

15B	CSE421: Computer Networks Quiz #1 Time: 15 Minutes	Marks: /15
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- Q1** The web server receives the photo upload request from Alice's PC. Below is a list of steps that the web server takes to process the received data. Mention the OSI layers associated with each of these tasks and **re-arrange** them in the correct order of the OSI model. **7**
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- A. The web application shows the uploaded photo.
 - B. The TCP layer reassembles the segments and ensures reliable delivery.
 - C. The sender's IP address is identified, and the packet is forwarded up the network stack.
 - D. The frames are checked for errors.
 - E. The received image file is decrypted and decompressed.
 - F. The web server extracts the data bits arriving through the physical medium.
 - G. The session between Alice's PC and the web server is maintained.
- Q2** Which TCP/IP layer is concerned with the **Physical address**, and what is the purpose of using this address **3**
- Q3** Identify the purpose of **synchronization** in the **session layer**. **3**

15A	CSE421: Computer Networks Quiz #2 Time: 20 Minutes	Marks: /15
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Q1 A university's document submission portal allows students to:

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- I. Remove an old assignment.
- II. Update a previously submitted assignment.
- III. Check if an assignment exists on the server without downloading it.

State HTTP methods should be used for each action.

Q2 Sajib is visiting nonte.shopping.com for the first time using his browser and added some items to his cart without logging in. A few days later, he again visited the same webpage and saw all the items were still present in his cart. **Identify how this is possible and how the website identifies whether a user has previously visited or not.**

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15B	CSE421: Computer Networks Quiz #2 Time: 20 Minutes	Marks: /15
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Q1 During interaction with a journal system, the following situations occur: **3**

- I. A researcher uploads a new paper, and the server signals that everything is fine so far and the upload should continue.
- II. A student tries to access a paper, and the server successfully returns the document.
- III. A professor bookmarks an old research paper's URL, but when they try to access it later, they're redirected to a new URL.

For each case above, identify the appropriate HTTP status code.

Q2 Your LAN utilisation is 90% whereas your access link utilisation is 15%. You are thinking of setting up a proxy server to handle the bottleneck created in your LAN. Explain how setting up a proxy server will affect the above scenario. **4**